

English Version

Problem 3.8 Bayesian estimation for a Bernoulli distribution

(a) Likelihood function

Given i.i.d. samples $\mathcal{D} = \{x_1, \dots, x_n\}$ from a Bernoulli distribution with parameter π , the likelihood is:

$$p(\mathcal{D}|\pi) = \prod_{i=1}^n p(x_i|\pi) = \prod_{i=1}^n \pi^{x_i} (1 - \pi)^{1-x_i} = \pi^{\sum_{i=1}^n x_i} (1 - \pi)^{n - \sum_{i=1}^n x_i} = \pi^s (1 - \pi)^{n-s}$$

where $s = \sum_{i=1}^n x_i$ is the sufficient statistic.

(b) Posterior distribution with uniform prior

Assume a uniform prior $p(\pi) = 1$ for $0 \leq \pi \leq 1$. The posterior is proportional to the likelihood times the prior:

$$p(\pi|\mathcal{D}) \propto p(\mathcal{D}|\pi)p(\pi) = \pi^s (1 - \pi)^{n-s}$$

The normalization constant is given by the integral:

$$\int_0^1 \pi^s (1 - \pi)^{n-s} d\pi = \frac{\Gamma(s+1)\Gamma(n-s+1)}{\Gamma(n+2)}$$

using the identity provided. Thus, the posterior distribution is:

$$p(\pi|\mathcal{D}) = \frac{\Gamma(n+2)}{\Gamma(s+1)\Gamma(n-s+1)} \pi^s (1 - \pi)^{n-s}$$

This is a Beta distribution with parameters $\alpha = s + 1$ and $\beta = n - s + 1$.

For $n = 1$, s can be 0 or 1:

- If $s = 0$, $p(\pi|\mathcal{D}) = \frac{\Gamma(3)}{\Gamma(1)\Gamma(2)} \pi^0 (1 - \pi)^1 = \frac{2!}{0!1!} (1 - \pi) = 2(1 - \pi)$
- If $s = 1$, $p(\pi|\mathcal{D}) = \frac{\Gamma(3)}{\Gamma(2)\Gamma(1)} \pi^1 (1 - \pi)^0 = \frac{2}{1 \cdot 1} \pi = 2\pi$

The plots for $n = 1$:

- For $s = 0$: a linear decreasing function from 2 at $\pi = 0$ to 0 at $\pi = 1$.

- For $s = 1$: a linear increasing function from 0 at $\pi = 0$ to 2 at $\pi = 1$.

(c) Predictive distribution

The predictive distribution for a new sample x is:

$$p(x|\mathcal{D}) = \int_0^1 p(x|\pi)p(\pi|\mathcal{D})d\pi$$

For $x = 1$:

$$p(x = 1|\mathcal{D}) = \int_0^1 \pi \cdot p(\pi|\mathcal{D})d\pi$$

Since $p(\pi|\mathcal{D})$ is Beta($s + 1, n - s + 1$), the mean is:

$$E[\pi|\mathcal{D}] = \frac{s + 1}{(s + 1) + (n - s + 1)} = \frac{s + 1}{n + 2}$$

Thus, $p(x = 1|\mathcal{D}) = \frac{s+1}{n+2}$.

The effective Bayesian estimate of π is the posterior mean $\frac{s+1}{n+2}$. This can be interpreted as adding two virtual samples, one success and one failure, to the ML estimate. This is known as Laplace smoothing or add-one smoothing.

(d) ML and MAP estimates with uniform prior

The ML estimate maximizes the likelihood $p(\mathcal{D}|\pi)$:

$$\hat{\pi}_{ML} = \frac{s}{n}$$

With a uniform prior, the MAP estimate maximizes the posterior, which is proportional to the likelihood, so:

$$\hat{\pi}_{MAP} = \frac{s}{n}$$

The advantage of the Bayesian estimate $\frac{s+1}{n+2}$ over the ML estimate is when n is small. For example, if $n = 1$ and $s = 0$, $\hat{\pi}_{ML} = 0$, which predicts no success ever, but the Bayesian estimate $\frac{1}{3}$ is more reasonable. The uniform prior adds virtual samples, making the estimate more robust to small sample sizes.

(e) MAP estimates with alternative priors

Consider priors $p_1(\pi) = 2\pi$ and $p_0(\pi) = 2(1 - \pi)$ for $0 \leq \pi \leq 1$.

For $p_1(\pi) = 2\pi$:

The posterior is:

$$p(\pi|\mathcal{D}) \propto p(\mathcal{D}|\pi)p_1(\pi) = \pi^s(1 - \pi)^{n-s} \cdot 2\pi = 2\pi^{s+1}(1 - \pi)^{n-s}$$

Maximizing the log-posterior:

$$\frac{d}{d\pi} [(s+1)\log \pi + (n-s)\log(1 - \pi)] = \frac{s+1}{\pi} - \frac{n-s}{1-\pi} = 0$$

Solving:

$$\frac{s+1}{\pi} = \frac{n-s}{1-\pi} \implies (s+1)(1-\pi) = (n-s)\pi \implies s+1 = \pi(n+1) \implies \hat{\pi}_{MAP} = \frac{s+1}{n+1}$$

For $p_0(\pi) = 2(1 - \pi)$:

The posterior is:

$$p(\pi|\mathcal{D}) \propto \pi^s(1 - \pi)^{n-s} \cdot 2(1 - \pi) = 2\pi^s(1 - \pi)^{n-s+1}$$

Maximizing the log-posterior:

$$\frac{d}{d\pi} [s\log \pi + (n-s+1)\log(1 - \pi)] = \frac{s}{\pi} - \frac{n-s+1}{1-\pi} = 0$$

Solving:

$$\frac{s}{\pi} = \frac{n-s+1}{1-\pi} \implies s(1-\pi) = (n-s+1)\pi \implies s = \pi(n+1) \implies \hat{\pi}_{MAP} = \frac{s}{n+1}$$

The effective Bayesian estimates:

- For p_1 , $\hat{\pi}_{MAP} = \frac{s+1}{n+1}$ adds one virtual success.
- For p_0 , $\hat{\pi}_{MAP} = \frac{s}{n+1}$ adds one virtual failure.